YEAR 10 PHYSICS, AQA GCSE, LESSON 6 OF 8

PARTICLE MODEL OF MATTER

LESSON 06

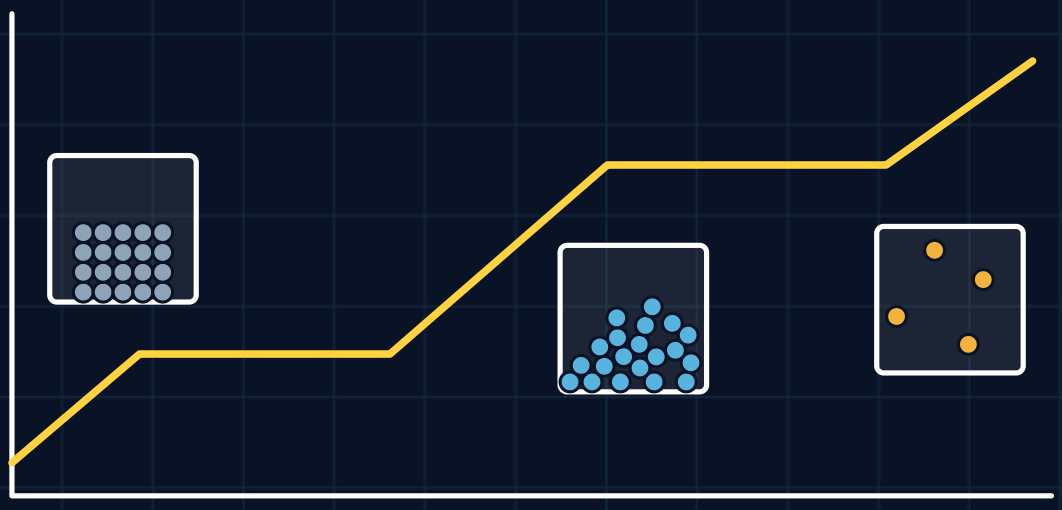
SPECIFIC LATENT HEAT

A scald from steam at 100 °C is far worse than a scald from water at 100 °C. Why?

IN THIS LESSON YOU WILL

- state what specific latent heat means
- calculate with $E = m\,L$
- tell fusion from vaporisation
- describe an experiment to measure the specific latent heat of fusion of water

ANSWER EDITION FOR TEACHERS AND SELF-MARKING



NAME _____ CLASS _____

START HERE

HOW THIS BOOKLET WORKS

This is the booklet for **Lesson 6** of the Particle Model of Matter unit. It builds on the lessons before it; the “Do now” questions and the quick answers at the back help you check what you remember. Keep the booklet: it is your classwork book *and* your revision guide.

DO NOW
Quick questions from earlier lessons. Remembering again is what makes it stick.

LEARN IT
The explanation and diagrams. Fill in each example from the slides with your teacher, then try “Your turn” alone. Read this again when you revise.

TRY IT
Practice that builds up: first with support, then on your own, then a challenge. Every second lesson ends with the Quick check from the slides.

PRACTICAL
Record your measurements and work out results from them.

EXAM CORNER
AQA-style questions with a tip on how the marks are given.

REVISE IT
Summary, cover-and-answer questions and an “I can” check to colour in.

! **Two kinds of question to know.** When a sentence has a gap, the choices are printed straight after it, like this: The particles in a solid are close together **(close together / far apart)**. When you see a box of choices, **draw a ring around** the correct one, as AQA exam papers ask. The unit of mass is **the newton** | **the kilogram** | **the joule** .

THE ROUTE THROUGH THE UNIT (THIS LESSON IS HIGHLIGHTED)

01 Solids, liquids and gases

02 Gas particles and pressure

03 Density

04 Measuring density (required practical)

05 Internal energy and heating curves

06 **Specific latent heat**
Latent heat, $E = m\,L$, fusion and vaporisation, measuring L for melting ice (slides 29–34)

07 Specific heat capacity

08 Measuring specific heat capacity (required practical)

The **equation toolkit** near the back has every equation you have met so far, with formula triangles to help you rearrange. Teaching deck: **panphy.app/gcsephy/decks/particle-**

model

Lesson

06 SPECIFIC LATENT HEAT

AQA Physics 4.3.2.3; Combined Science 6.3.2.3

BIG QUESTION **A scald from steam at 100 °C is far worse than a scald from water at 100 °C. Why?**

BY THE END YOU CAN

- state what specific latent heat means
- calculate with $E = m\,L$
- tell fusion from vaporisation
- describe an experiment to measure the specific latent heat of fusion of water

latent heat · **specific latent heat** · **fusion** · **vaporisation** · **joule per kilogram (J/kg)**

YOU WILL NEED

Per group, or as a demonstration: two funnels of crushed ice over beakers, 12 V immersion heater, joulemeter, power supply, balance, stopwatch.

Teaching deck, slides 29–34
panphy.app/gcsephy/decks/particle-model/#29

REMEMBER

DO NOW

From earlier lessons. No notes!

1
Define internal energy.

The total kinetic and potential energy of all the particles in a system.

2
What happens to the temperature of a substance while it changes state?

It stays constant.

3
Convert 250 g to kilograms.

0.250 kg

EXPLAIN

THE HIDDEN ENERGY

The energy needed for a substance to change state is called **latent heat**. “Latent” means hidden: the energy goes in, the internal energy increases, but a thermometer shows no change.

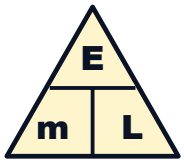
! The **specific latent heat** of a substance is the amount of energy required to change the state of **one kilogram** of the substance with **no change in temperature**.

$$E = m\,L$$

energy for a change of state = mass × specific latent heat

E in J, m in kg, L in J/kg

Rearranged: $L = \frac{E}{m}$ and $m = \frac{E}{L}$



TWO KINDS

	CHANGE OF STATE	VALUE FOR WATER
Specific latent heat of fusion	solid to liquid (melting), or liquid to solid (freezing)	334 000 J/kg
Specific latent heat of vaporisation	liquid to vapour (boiling), or vapour to liquid (condensing)	2 260 000 J/kg

Boiling needs much more energy than melting, because the particles must be pulled completely apart. When the change goes the other way, the same amount of energy is **given out**. Steam that condenses on your skin transfers 2 260 000 J for every kilogram, *before* it starts to cool as hot water. That is why steam scalds so badly.

WITH YOUR TEACHER — SLIDE 31

Calculate the energy needed to melt 0.50 kg of ice at 0 °C. Specific latent heat of fusion of water = 334 000 J/kg

EQUATION $E = m\,L$

SUBSTITUTE $E = 0.50 \times 334\,000$

ANSWER $E = 167\,000$ J

YOUR TURN

Calculate the energy needed to boil away 0.20 kg of water at 100 °C. Specific latent heat of vaporisation of water = 2 260 000 J/kg

EQUATION $E = m\,L$

SUBSTITUTE $E = 0.20 \times 2\,260\,000$

ANSWER $E = 452\,000$ J

WITH YOUR TEACHER SLIDE 31

A heater transfers 90 000 J to a block of wax at its melting point. The specific latent heat of fusion of the wax is 150 000 J/kg. Calculate the mass of wax that melts.

EQUATION	$E = m\,L$
REARRANGE	$m = \frac{E}{L}$
SUBSTITUTE	$m = \frac{90\,000}{150\,000}$
ANSWER	**m = 0.60 kg**

YOUR TURN

It takes 50 400 J to melt 0.40 kg of a metal at its melting point. Calculate the specific latent heat of fusion of the metal.

EQUATION	$E = m\,L$
REARRANGE	$L = \frac{E}{m}$
SUBSTITUTE	$L = \frac{50\,400}{0.40}$
ANSWER	$L = 126\,000$ J/kg

Get the marks: There is no temperature in this equation. If a question gives a temperature change, you need a different equation (Lesson 7).

PRACTISE

TRY IT

1 Draw a ring around the correct answer in each box.

The unit of specific latent heat is [J | (J/kg) | J/kg °C]

Ice melting uses the specific latent heat of [(fusion) | vaporisation]

Steam condensing uses the specific latent heat of [fusion | (vaporisation)]

2 SUPPORTED Calculate the energy needed to melt 1.5 kg of ice at 0 °C. Specific latent heat of fusion of water = 334 000 J/kg

EQUATION	$E = m\,L$
SUBSTITUTE	$E = 1.5 \times 334\,000$
ANSWER	**501 000 J** unit:

3 SUPPORTED A kettle boils away 150 g of water at 100 °C. Calculate the energy transferred to the water. Specific latent heat of vaporisation of water = 2 260 000 J/kg

Convert the mass first!

CONVERT	**m = 150 ÷ 1000 = 0.150 kg**
EQUATION	$E = m\,L$
SUBSTITUTE	$E = 0.150 \times 2\,260\,000$
ANSWER	**339 000 J** unit:

4 Complete the table. Show your working underneath.

Change of state	Mass / kg	Specific latent heat / J/kg	Energy / J
Lead melting	**2.0**	**23 000**	**46 000**
Ethanol boiling	**0.50**	**850 000**	**425 000**
Aluminium melting	**0.50**	**400 000**	**200 000**

5 CHALLENGE A freezer removes 66 800 J from water at 0 °C. Calculate the mass of ice that forms. Specific latent heat of fusion of water = 334 000 J/kg

$$m = \frac{E}{L} = \frac{66\,800}{334\,000} = 0.20 \text{ kg}$$

INVESTIGATE

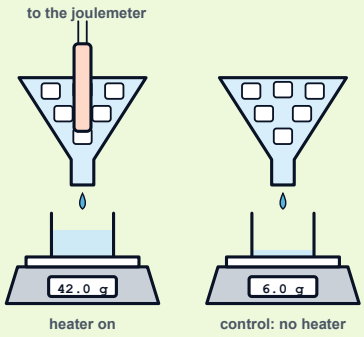
PRACTICAL: MELTING ICE

Measure the specific latent heat of fusion of water.

STAY SAFE

- The heater gets hot: switch it on only when it is surrounded by ice.
- Keep water away from the power supply and wipe up spills.

1. Fill two funnels with the same mass of crushed ice. Put a beaker under each.
2. Put the heater in one funnel. The other is the **control**: it shows how much ice melts just because the room is warm.
3. Switch on the heater and the joulemeter for 5 minutes.
4. Measure the mass of water in each beaker.
5. Mass melted by the heater = mass from the heated funnel − mass from the control.



6. Record your results.

Measurement	Value
Energy supplied (joulemeter) / J	**12 000**
Mass of water from the heated funnel / g	**42.0**
Mass of water from the control funnel / g	**6.0**
Mass melted by the heater / kg	**0.0360**

7. Calculate the specific latent heat of fusion of water from your results. Use $E = m\,L$

EQUATION	$E = m\,L$
REARRANGE	$L = \frac{E}{m}$
SUBSTITUTE	$L = \frac{12\,000}{0.0360}$
ANSWER	**333 000 J/kg** unit:

8. Explain why the control funnel is needed.

Some ice melts because energy is transferred from the warm room. The control measures this, so it can be subtracted to find the mass melted by the heater alone.

CHECKPOINT

QUICK CHECK

Slide 34, Lessons 5–6. Answer each question, then check it as your teacher reveals the answer.

1

Internal energy is the total of which two energies?

The kinetic energy and potential energy of the particles.

2

What does a flat section of a heating graph show?

A change of state.

3

Which energy of the particles increases while ice melts?

Their potential energy.

4

State the unit of specific latent heat.

J/kg

5

Calculate the energy needed to melt 2.0 kg of ice. L = 334 000 J/kg

$E = 2.0 \times 334\,000 = 668\,000$ **J**

6

Liquid to vapour: latent heat of fusion or of vaporisation?

Vaporisation.

AQA-STYLE

EXAM CORNER

Write in full sentences. Show every step of a calculation.

Get the marks: “Calculate” with a mass in grams: convert to kilograms before you substitute.

9 A 60 g ice cube at 0 °C melts in a drink. Calculate the energy transferred to the ice cube as it melts. Specific latent heat of fusion of water = 334 000 J/kg **[3 marks]**

CONVERT	**m = 0.060 kg (1)**
EQUATION	$E = m\,L$
SUBSTITUTE	$E = 0.060 \times 334\,000$ **(1)**
ANSWER	**20 040 J (accept 20 000 J) (1)** unit:

10 State what is meant by the specific latent heat of vaporisation of a substance. **[2 marks]**

The energy needed to change 1 kg of the substance from liquid to vapour (1) with no change in temperature (1).

REVIEW LATER

REVISE IT

Use this box after the lesson and before tests.

SUMMARY

- Latent heat is the energy needed for a change of state. The temperature does not change.
- Specific latent heat L: the energy to change the state of 1 kg, in J/kg.
- $E = m\,L$; rearranged, $L = \dfrac{E}{m}$ and $m = \dfrac{E}{L}$.
- Fusion: solid and liquid. Vaporisation: liquid and vapour.
- The same energy is given out when the change is reversed (freezing, condensing).

COVER THE PAGE AND ANSWER

1 Define specific latent heat.

2 Write the equation for the energy for a change of state.

3 State the unit of specific latent heat.

4 Which change of state does “fusion” describe?

I CAN…

- define specific latent heat
- calculate energy, mass or L with $E = m\,L$
- tell fusion from vaporisation
- describe how to measure the specific latent heat of fusion of water

REFERENCE

EQUATION TOOLKIT

Every equation from Lessons 1–6. AQA prints all of them on the equation sheet in the exam, but you still need to know what each symbol means, its unit, and how to use it.

1. Cover the quantity you want to find.
2. What is left side by side? **Multiply**.
3. One above the other? **Divide** the top by the bottom.

Example: to find V from $\rho = \frac{m}{V}$, cover V. m is above ρ, so $V = \frac{m}{\rho}$.

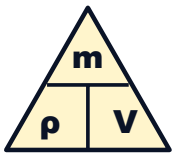
$$\rho = \frac{m}{V}$$

Lesson 3

density = mass ÷ volume

kg/m³, kg, m³ (or g/cm³, g, cm³)

$m = \rho V$ and $V = \frac{m}{\rho}$



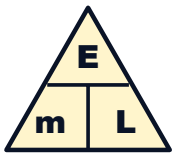
$$E = m\,L$$

Lesson 6

energy for a change of state = mass × specific latent heat

J, kg, J/kg

$L = \frac{E}{m}$ and $m = \frac{E}{L}$



1
Write what you know. Convert first: g → kg, cm³ → m³, kJ → J.

2
Write the equation in symbols.

3
Rearrange if you need to, then substitute the numbers.

4
Give the answer with a unit. Ask: is it sensible?

REFERENCE

KEY WORDS FROM THIS LESSON

Learn the meaning and the spelling. In exams, precise words earn marks: say “the particles vibrate about fixed positions”, not “the particles shake”.

Latent heat: The energy needed for a substance to change state, with no change in temperature

Specific latent heat (L): The energy needed to change the state of 1 kg of a substance with no change in temperature, in J/kg

Fusion: Melting: the change from solid to liquid

Vaporisation: The change from liquid to vapour

CHECK YOURSELF

QUICK ANSWERS

Answers to the “Do now” and “Cover the page” questions, so you can check yourself when revising. Answers to the longer questions are in the answer edition.

DO NOW

1. The total kinetic and potential energy of all the particles in a system.
2. It stays constant.
3. 0.250 kg

COVER THE PAGE

1. The energy needed to change the state of 1 kg of a substance with no change in temperature.
2. $E = m L$
3. J/kg
4. Solid to liquid (melting).